

NAME: A.SANJAY KUMAR

Email : attellisanjay29@gmail.com

Date: 29-12-2025 a Day 2 – Querying & Modifying Data

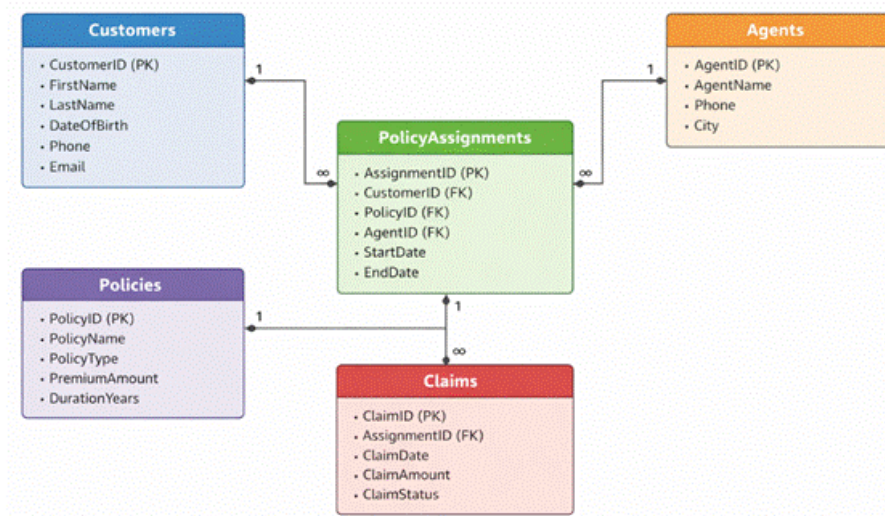
Create a Sample Insurance database named **InsuranceDB** with tables, constraints, and initial sample data. Based on following a diagram and descriptions.

Entities we'll model:

- **Customers** – people who buy insurance
- **Policies** – insurance products (Health, Motor, Life)
- **Agents** – insurance agents
- **PolicyAssignments** – which customer bought which policy
- **Claims** – claims raised against policies

Relationship Explanation

- **One Customer → Many PolicyAssignments**
- **One Policy → Many PolicyAssignments**
- **One Agent → Many PolicyAssignments**
- **One PolicyAssignment → Many Claims**



1. Create Database command

```
CREATE DATABASE insurance_db;
```

Commands completed successfully.

Completion time: 2025-12-29T16:57:25.4074763+05:30

2. Create table commands for all tables

```
CREATE TABLE insurance_customers (  
  customer_id INT PRIMARY KEY,  
  first_name VARCHAR(50),  
  last_name VARCHAR(50),  
  dob DATE,  
  phone VARCHAR(15),  
  email VARCHAR(100),  
  created_at DATE,  
  updated_at DATE  
);
```

Output:

	Column Name	Data Type	Allow Nulls
🔍	customer_id	int	☐
	first_name	varchar(50)	☐
	last_name	varchar(50)	☐
	dob	date	☐
	phone	varchar(15)	☒
	email	varchar(100)	☒
	created_at	date	☒
	updated_at	date	☒
			☐

```
CREATE TABLE policies (  
  policy_id INT PRIMARY KEY,  
  policy_name VARCHAR(50),  
  policy_type VARCHAR(50),  
  pre_amount INT,  
  duration_year INT,  
  created_at DATE,  
  updated_at DATE  
);
```

Output:

	Column Name	Data Type	Allow Nulls
🔑	policy_id	int	☐
	policy_name	varchar(50)	☐
	policy_type	varchar(50)	☐
	pre_amount	int	☐
	duration_year	int	☐
	created_at	date	☒
	updated_at	date	☒
			☐

```
CREATE TABLE agents (
  agent_id INT PRIMARY KEY,
  agent_name VARCHAR(100),
  phone VARCHAR(15),
  city VARCHAR(50),
  created_at DATE,
  updated_at DATE
```

```
);
```

Output:

	Column Name	Data Type	Allow Nulls
🔑	agent_id	int	☐
	agent_name	varchar(100)	☐
	phone	varchar(15)	☒
	city	varchar(50)	☒
	created_at	date	☒
	updated_at	date	☒
			☐

```
CREATE TABLE policy_assignments (
  assignment_id INT PRIMARY KEY,
  customer_id INT,
  policy_id INT,
  agent_id INT,
  start_date DATE,
  end_date DATE,
  created_at DATE,
  updated_at DATE,
  FOREIGN KEY (customer_id) REFERENCES insurance_customers(customer_id),
  FOREIGN KEY (policy_id) REFERENCES policies(policy_id),
  FOREIGN KEY (agent_id) REFERENCES agents(agent_id)
```

```
);
```

Output:

	Column Name	Data Type	Allow Nulls
🔑	assignment_id	int	☐
	customer_id	int	☒
	policy_id	int	☒
	agent_id	int	☒
	start_date	date	☒
	end_date	date	☒
	created_at	date	☒
	updated_at	date	☒
			☐

```
CREATE TABLE claims (
  claim_id INT PRIMARY KEY,
  assignment_id INT,
  claim_date DATE,
  claim_amount INT,
  claim_status VARCHAR(30),
  created_at DATE,
  updated_at DATE,
  FOREIGN KEY (assignment_id) REFERENCES policy_assignments(assignment_id)
);
```

	Column Name	Data Type	Allow Nulls
🔑	claim_id	int	☐
	assignment_id	int	☒
	claim_date	date	☐
	claims_amount	int	☐
	claim_status	varchar(30)	☐
	created_at	date	☒
	updated_at	date	☒
			☐

3. Insert commands for all tables

Data Inserted in to insurance_customers Table:

```
INSERT INTO insurance_customers VALUES
(1,'Sanjay','Attelli','2005-01-29','8919200290','sanjay.attelli@gmail.com','2024-01-05','2024-06-10'),
(2,'Manoj','Kumar','2004-05-25','9123456789','manoj.kumar@yahoo.com','2024-02-12','2024-07-01'),
(3,'Amit','Sharma','1998-04-12','9000000001','amit.sharma@gmail.com','2024-03-01','2024-06-01'),
(4,'Priya','Singh','1996-09-21','9000000002','priya.singh@gmail.com','2024-03-02','2024-06-02'),
(5,'Rahul','Verma','1994-01-15','9000000003','rahul.verma@yahoo.com','2024-03-03','2024-06-03'),
(6,'Neha','Gupta','2000-11-30','9000000004','neha.gupta@gmail.com','2024-03-04','2024-06-04'),
(7,'Vikas','Reddy','1992-07-18','9000000005','vikas.reddy@gmail.com','2024-03-05','2024-06-05'),
(8,'Pooja','Nair','1999-02-25','9000000006','pooja.nair@gmail.com','2024-03-06','2024-06-06'),
(9,'Arjun','Patel','1991-12-09','9000000007','arjun.patel@yahoo.com','2024-03-07','2024-06-07'),
(10,'Kiran','Das','1997-06-14','9000000008','kiran.das@gmail.com','2024-03-08','2024-06-08'),
(11,'Sneha','Joshi','1995-10-10','9000000009','sneha.joshi@gmail.com','2024-03-09','2024-06-09'),
(12,'Rohit','Mehta','1993-05-05','9000000010','rohit.mehta@gmail.com','2024-03-10','2024-06-10');
```

Results		Messages						
	customer_id	first_name	last_name	dob	phone	email	created_at	updated_at
1	1	Sanjay	Atelli	2005-01-29	8919200290	sanjay.attelli@gmail.com	2024-01-05	2024-06-10
2	2	manoj	Kumar	2004-05-25	9123456789	manoj.kumar@yahoo.com	2024-02-12	2024-07-01
3	3	Amit	Sharma	1998-04-12	90000000001	amit.sharma@gmail.com	2024-03-01	2024-06-01
4	4	Priya	Singh	1996-09-21	90000000002	priya.singh@gmail.com	2024-03-02	2024-06-02
5	5	Rahul	Verma	1994-01-15	90000000003	rahul.verma@yahoo.com	2024-03-03	2024-06-03
6	6	Neha	Gupta	2000-11-30	90000000004	neha.gupta@gmail.com	2024-03-04	2024-06-04
7	7	Vikas	Reddy	1992-07-18	90000000005	vikas.reddy@gmail.com	2024-03-05	2024-06-05
8	8	Pooja	Nair	1999-02-25	90000000006	pooja.nair@gmail.com	2024-03-06	2024-06-06
9	9	Arjun	Patel	1991-12-09	90000000007	arjun.patel@yahoo.com	2024-03-07	2024-06-07
10	10	Kiran	Das	1997-06-14	90000000008	kiran.das@gmail.com	2024-03-08	2024-06-08
11	11	Sneha	Joshi	1995-10-10	90000000009	sneha.joshi@gmail.com	2024-03-09	2024-06-09
12	12	Rohit	Mehta	1993-05-05	90000000010	rohit.mehta@gmail.com	2024-03-10	2024-06-10

Data Inserted in to PoliciesTable:

INSERT INTO policies VALUES

(101,'Life Secure Plan','Life Insurance',12000,20,'2023-12-01','2024-05-01'),
 (102,'Health Plus Policy','Health Insurance',8500,1,'2024-01-15','2024-06-20'),
 (103,'Child Secure','Life Insurance',15000,18,'2024-02-01','2024-06-01'),
 (104,'Senior Health','Health Insurance',22000,1,'2024-02-02','2024-06-02'),
 (105,'Family Health Plus','Health Insurance',18000,2,'2024-02-03','2024-06-03'),
 (106,'Term Life Gold','Life Insurance',25000,30,'2024-02-04','2024-06-04'),
 (107,'Personal Accident','General Insurance',6000,1,'2024-02-05','2024-06-05'),
 (108,'Travel Secure','Travel Insurance',4000,1,'2024-02-06','2024-06-06'),
 (109,'Home Safe','Property Insurance',12000,10,'2024-02-07','2024-06-07'),
 (110,'Vehicle Protect','Motor Insurance',9000,1,'2024-02-08','2024-06-08'),
 (111,'Health Premium','Health Insurance',30000,3,'2024-02-09','2024-06-09'),
 (112,'Life Platinum','Life Insurance',35000,25,'2024-02-10','2024-06-10');

Results		Messages					
	policy_id	policy_name	policy_type	pre_amount	duration_year	created_at	updated_at
1	101	Life Secure Plan	Life Insurance	12000	20	2023-12-01	2024-05-01
2	102	Health Plus Policy	Health Insurance	8500	1	2024-01-15	2024-06-20
3	103	Child Secure	Life Insurance	15000	18	2024-02-01	2024-06-01
4	104	Senior Health	Health Insurance	22000	1	2024-02-02	2024-06-02
5	105	Family Health Plus	Health Insurance	18000	2	2024-02-03	2024-06-03
6	106	Term Life Gold	Life Insurance	25000	30	2024-02-04	2024-06-04
7	107	Personal Accident	General Insurance	6000	1	2024-02-05	2024-06-05
8	108	Travel Secure	Travel Insurance	4000	1	2024-02-06	2024-06-06
9	109	Home Safe	Property Insurance	12000	10	2024-02-07	2024-06-07
10	110	Vehicle Protect	Motor Insurance	9000	1	2024-02-08	2024-06-08
11	111	Health Premium	Health Insurance	30000	3	2024-02-09	2024-06-09
12	112	Life Platinum	Life Insurance	35000	25	2024-02-10	2024-06-10

Data Inserted in to agents Table:

```
INSERT INTO agents VALUES
(5001,'Sanjay Kumar','8919200290','Hyderabad','2023-11-10','2024-04-12'),
(5002,'Monoj Kumar','9123456781','Delhi','2023-10-05','2024-03-18'),
(5003,'Varshith','9988776651','Chennai','2023-09-20','2024-02-25'),
(5004,'Amit Rao','9011111111','Mumbai','2023-08-01','2024-04-01'),
(5005,'Priya Shah','9011111112','Pune','2023-08-02','2024-04-02'),
(5006,'Rakesh Jain','9011111113','Delhi','2023-08-03','2024-04-03'),
(5007,'Sunita Roy','9011111114','Kolkata','2023-08-04','2024-04-04'),
(5008,'Deepak Mishra','9011111115','Bhopal','2023-08-05','2024-04-05'),
(5009,'Anil Kumar','9011111116','Chennai','2023-08-06','2024-04-06'),
(5010,'Meena Iyer','9011111117','Coimbatore','2023-08-07','2024-04-07'),
(5011,'Suresh Naik','9011111118','Goa','2023-08-08','2024-04-08'),
(5012,'Kavya Shetty','9011111119','Mangalore','2023-08-09','2024-04-09'),
(5013,'Nitin Kulkarni','9011111120','Nagpur','2023-08-10','2024-04-10');
```

	agent_id	agent_name	phone	city	created_at	updated_at
1	5001	sanjay kumar	8919200290	Hyderabad	2023-11-10	2024-04-12
2	5002	monoj kumar	9123456781	delhi	2023-10-05	2024-03-18
3	5003	varshith	9988776651	Chennai	2023-09-20	2024-02-25
4	5004	Amit Rao	9011111111	Mumbai	2023-08-01	2024-04-01
5	5005	Priya Shah	9011111112	Pune	2023-08-02	2024-04-02
6	5006	Rakesh Jain	9011111113	Delhi	2023-08-03	2024-04-03
7	5007	Sunita Roy	9011111114	Kolkata	2023-08-04	2024-04-04
8	5008	Deepak Mishra	9011111115	Bhopal	2023-08-05	2024-04-05
9	5009	Anil Kumar	9011111116	Chennai	2023-08-06	2024-04-06
10	5010	Meena Iyer	9011111117	Coimbatore	2023-08-07	2024-04-07
11	5011	Suresh Naik	9011111118	Goa	2023-08-08	2024-04-08
12	5012	Kavya Shetty	9011111119	Mangalore	2023-08-09	2024-04-09
13	5013	Nitin Kulkarni	9011111120	Nagpur	2023-08-10	2024-04-10

Data Inserted in to policy_assignments Table:

```
INSERT INTO policy_assignments VALUES
(1,1,101,5001,'2024-01-01','2044-01-01','2024-01-02','2024-06-01'),
(2,2,102,5002,'2024-06-15','2025-06-15','2024-06-16','2024-07-10'),
(3,3,103,5004,'2024-01-10','2042-01-10','2024-01-11','2024-06-01'),
(4,4,104,5005,'2024-02-01','2025-02-01','2024-02-02','2024-06-02'),
(5,5,105,5006,'2024-03-01','2026-03-01','2024-03-02','2024-06-03'),
(6,6,106,5007,'2024-04-01','2054-04-01','2024-04-02','2024-06-04'),
(7,7,107,5008,'2024-05-01','2025-05-01','2024-05-02','2024-06-05'),
(8,8,108,5009,'2024-06-01','2025-06-01','2024-06-02','2024-06-06'),
(9,9,109,5010,'2024-07-01','2034-07-01','2024-07-02','2024-06-07'),
(10,10,110,5011,'2024-08-01','2025-08-01','2024-08-02','2024-06-08'),
(11,11,111,5012,'2024-09-01','2027-09-01','2024-09-02','2024-06-09'),
(12,12,112,5013,'2024-10-01','2049-10-01','2024-10-02','2024-06-10');
```

Results		Messages						
	assignment_id	customer_id	policy_id	agent_id	start_date	end_date	created_at	updated_at
1	1	1	101	5001	2024-01-01	2044-01-01	2024-01-02	2024-06-01
2	2	2	102	5002	2024-06-15	2025-06-15	2024-06-16	2024-07-10
3	3	3	103	5004	2024-01-10	2042-01-10	2024-01-11	2024-06-01
4	4	4	104	5005	2024-02-01	2025-02-01	2024-02-02	2024-06-02
5	5	5	105	5006	2024-03-01	2026-03-01	2024-03-02	2024-06-03
6	6	6	106	5007	2024-04-01	2054-04-01	2024-04-02	2024-06-04
7	7	7	107	5008	2024-05-01	2025-05-01	2024-05-02	2024-06-05
8	8	8	108	5009	2024-06-01	2025-06-01	2024-06-02	2024-06-06
9	9	9	109	5010	2024-07-01	2034-07-01	2024-07-02	2024-06-07
10	10	10	110	5011	2024-08-01	2025-08-01	2024-08-02	2024-06-08
11	11	11	111	5012	2024-09-01	2027-09-01	2024-09-02	2024-06-09
12	12	12	112	5013	2024-10-01	2049-10-01	2024-10-02	2024-06-10

Data Inserted in to claims Table:

INSERT INTO claims VALUES

(1,1,'2024-03-10',50000,'Approved','2024-03-11','2024-03-20'),
 (2,2,'2024-08-05',12000,'Pending','2024-08-06','2024-08-15'),
 (3,3,'2024-04-15',20000,'Approved','2024-04-16','2024-04-20'),
 (4,4,'2024-05-10',15000,'Pending','2024-05-11','2024-05-18'),
 (5,5,'2024-06-05',30000,'Rejected','2024-06-06','2024-06-12'),
 (6,6,'2024-07-01',50000,'Approved','2024-07-02','2024-07-10'),
 (7,7,'2024-08-12',8000,'Pending','2024-08-13','2024-08-20'),
 (8,8,'2024-09-18',12000,'Approved','2024-09-19','2024-09-25'),
 (9,9,'2024-10-22',25000,'Approved','2024-10-23','2024-10-30'),
 (10,10,'2024-11-05',9000,'Rejected','2024-11-06','2024-11-12'),
 (11,11,'2024-11-20',40000,'Pending','2024-11-21','2024-11-28'),
 (12,12,'2024-12-01',60000,'Approved','2024-12-02','2024-12-10');

	claim_id	assignment_id	claim_date	claims_amount	claim_status	created_at	updated_at
1	1	1	2024-03-10	50000	Approved	2024-03-11	2024-03-20
2	2	2	2024-08-05	12000	Pending	2024-08-06	2024-08-15
3	3	3	2024-04-15	20000	Approved	2024-04-16	2024-04-20
4	4	4	2024-05-10	15000	Pending	2024-05-11	2024-05-18
5	5	5	2024-06-05	30000	Rejected	2024-06-06	2024-06-12
6	6	6	2024-07-01	50000	Approved	2024-07-02	2024-07-10
7	7	7	2024-08-12	8000	Pending	2024-08-13	2024-08-20
8	8	8	2024-09-18	12000	Approved	2024-09-19	2024-09-25
9	9	9	2024-10-22	25000	Approved	2024-10-23	2024-10-30
10	10	10	2024-11-05	9000	Rejected	2024-11-06	2024-11-12
11	11	11	2024-11-20	40000	Pending	2024-11-21	2024-11-28
12	12	12	2024-12-01	60000	Approved	2024-12-02	2024-12-10

4. View all records from insurance_customers

```
SELECT * FROM insurance_customers;
```

Results

Messages

	customer_id	first_name	last_name	dob	phone	email	created_at	updated_at
1	1	Sanjay	Attelli	2005-01-29	8919200290	sanjay.attelli@gmail.com	2024-01-05	2024-06-10
2	2	manoj	Kumar	2004-05-25	9123456789	manoj.kumar@yahoo.com	2024-02-12	2024-07-01
3	3	Amit	Sharma	1998-04-12	9000000001	amit.sharma@gmail.com	2024-03-01	2024-06-01
4	4	Priya	Singh	1996-09-21	9000000002	priya.singh@gmail.com	2024-03-02	2024-06-02
5	5	Rahul	Verma	1994-01-15	9000000003	rahul.verma@yahoo.com	2024-03-03	2024-06-03
6	6	Neha	Gupta	2000-11-30	9000000004	neha.gupta@gmail.com	2024-03-04	2024-06-04
7	7	Vikas	Reddy	1992-07-18	9000000005	vikas.reddy@gmail.com	2024-03-05	2024-06-05
8	8	Pooja	Nair	1999-02-25	9000000006	pooja.nair@gmail.com	2024-03-06	2024-06-06
9	9	Arjun	Patel	1991-12-09	9000000007	arjun.patel@yahoo.com	2024-03-07	2024-06-07
10	10	Kiran	Das	1997-06-14	9000000008	kiran.das@gmail.com	2024-03-08	2024-06-08
11	11	Sneha	Joshi	1995-10-10	9000000009	sneha.joshi@gmail.com	2024-03-09	2024-06-09
12	12	Rohit	Mehta	1993-05-05	9000000010	rohit.mehta@gmail.com	2024-03-10	2024-06-10

5. View PolicyAssignments with CustomerId, PolicyId, StartDate, EndDate

```
SELECT customer_id, policy_id, start_date, end_date  
FROM policy_assignments;
```

Results

Messages

	customer_id	policy_id	start_date	end_date
1	1	101	2024-01-01	2044-01-01
2	2	102	2024-06-15	2025-06-15
3	3	103	2024-01-10	2042-01-10
4	4	104	2024-02-01	2025-02-01
5	5	105	2024-03-01	2026-03-01
6	6	106	2024-04-01	2054-04-01
7	7	107	2024-05-01	2025-05-01
8	8	108	2024-06-01	2025-06-01
9	9	109	2024-07-01	2034-07-01
10	10	110	2024-08-01	2025-08-01
11	11	111	2024-09-01	2027-09-01
12	12	112	2024-10-01	2049-10-01

6. Display all policies of Health type

```
SELECT * FROM policies
WHERE policy_type = 'Health Insurance';
```

Results

Messages

	policy_id	policy_name	policy_type	pre_amount	duration_year	created_at	updated_at
1	102	Health Plus Policy	Health Insurance	8500	1	2024-01-15	2024-06-20
2	104	Senior Health	Health Insurance	22000	1	2024-02-02	2024-06-02
3	105	Family Health Plus	Health Insurance	18000	2	2024-02-03	2024-06-03
4	111	Health Premium	Health Insurance	30000	3	2024-02-09	2024-06-09

7. Premium > 10000 and DurationYears = 1

```
SELECT * FROM policies
WHERE pre_amount > 10000 AND duration_year = 1;
```

Results

Messages

	policy_id	policy_name	policy_type	pre_amount	duration_year	created_at	updated_at
1	104	Senior Health	Health Insurance	22000	1	2024-02-02	2024-06-02

Click to select all grid cells

8. Display unique agent city names

```
SELECT DISTINCT city FROM agents;
```

Results		Messages	
	city		
1	Bhopal		
2	Chennai		
3	Coimbatore		
4	delhi		
5	Goa		
6	Hyderabad		
7	Kolkata		
8	Mangalore		
9	Mumbai		
10	Nagpur		
11	Pune		

9. Policies using OR

```
SELECT * FROM policies
WHERE policy_type = 'Life Insurance'
OR policy_type = 'Health Insurance'
OR policy_type = 'Motor Insurance';
```

Results

Messages

	policy_id	policy_name	policy_type	pre_amount	duration_year	created_at	updated_at
1	101	Life Secure Plan	Life Insurance	12000	20	2023-12-01	2024-05-01
2	102	Health Plus Policy	Health Insurance	8500	1	2024-01-15	2024-06-20
3	103	Child Secure	Life Insurance	15000	18	2024-02-01	2024-06-01
4	104	Senior Health	Health Insurance	22000	1	2024-02-02	2024-06-02
5	105	Family Health Plus	Health Insurance	18000	2	2024-02-03	2024-06-03
6	106	Term Life Gold	Life Insurance	25000	30	2024-02-04	2024-06-04
7	110	Vehicle Protect	Motor Insurance	9000	1	2024-02-08	2024-06-08
8	111	Health Premium	Health Insurance	30000	3	2024-02-09	2024-06-09
9	112	Life Platinum	Life Insurance	35000	25	2024-02-10	2024-06-10

10. Policies using IN

```
SELECT * FROM policies
WHERE policy_type IN ('Life Insurance','Health Insurance','Motor Insurance');
```

Results

Messages

	policy_id	policy_name	policy_type	pre_amount	duration_year	created_at	updated_at
1	101	Life Secure Plan	Life Insurance	12000	20	2023-12-01	2024-05-01
2	102	Health Plus Policy	Health Insurance	8500	1	2024-01-15	2024-06-20
3	103	Child Secure	Life Insurance	15000	18	2024-02-01	2024-06-01
4	104	Senior Health	Health Insurance	22000	1	2024-02-02	2024-06-02
5	105	Family Health Plus	Health Insurance	18000	2	2024-02-03	2024-06-03
6	106	Term Life Gold	Life Insurance	25000	30	2024-02-04	2024-06-04
7	110	Vehicle Protect	Motor Insurance	9000	1	2024-02-08	2024-06-08
8	111	Health Premium	Health Insurance	30000	3	2024-02-09	2024-06-09
9	112	Life Platinum	Life Insurance	35000	25	2024-02-10	2024-06-10

11. Customers born using >= and <=

```
SELECT * FROM insurance_customers
WHERE dob >= '2001-01-01' AND dob <= '2020-12-31';
```

Results

Messages

	customer_id	first_name	last_name	dob	phone	email	created_at	updated_at
1	1	Sanjay	Attelli	2005-01-29	8919200290	sanjay.attelli@gmail.com	2024-01-05	2024-06-10
2	2	manoj	Kumar	2004-05-25	9123456789	manoj.kumar@yahoo.com	2024-02-12	2024-07-01

12. Customers born using BETWEEN

```
SELECT * FROM insurance_customers
WHERE dob BETWEEN '2001-01-01' AND '2020-12-31';
```

	customer_id	first_name	last_name	dob	phone	email	created_at	updated_at
1	1	Sanjay	Attelli	2005-01-29	8919200290	sanjay.attelli@gmail.com	2024-01-05	2024-06-10
2	2	manoj	Kumar	2004-05-25	9123456789	manoj.kumar@yahoo.com	2024-02-12	2024-07-01

13. Claims with Rejected status

```
SELECT * FROM claims
WHERE claim_status = 'Rejected';
```

	claim_id	assignment_id	claim_date	claims_amount	claim_status	created_at	updated_at
1	5	5	2024-06-05	30000	Rejected	2024-06-06	2024-06-12
2	10	10	2024-11-05	9000	Rejected	2024-11-06	2024-11-12

14. Agents staying in city whose second letter is 'a'

```
SELECT * FROM agents
WHERE city LIKE '_a%';
```

	agent_id	agent_name	phone	city	created_at	updated_at
1	5012	Kavya Shetty	9011111119	Mangalore	2023-08-09	2024-04-09
2	5013	Nitin Kulkarni	9011111120	Nagpur	2023-08-10	2024-04-10

15. Highest and Lowest claim amount

```
SELECT MIN(claim_amount) AS min_claim,
       MAX(claim_amount) AS max_claim
FROM claims;
```

	min_claim	max_claim
1	8000	60000

16. Latest claim record

```
SELECT TOP 1 *
FROM claims
ORDER BY claim_date DESC;
```

Results		Messages					
	claim_id	assignment_id	claim_date	claims_amount	claim_status	created_at	updated_at
1	12	12	2024-12-01	60000	Approved	2024-12-02	2024-12-10

17. Increase Health policy premium by 10%

UPDATE policies

SET pre_amount = pre_amount * 1.10

WHERE policy_type = 'Health Insurance';

Messages	
(4 rows affected)	
Completion time: 2025-12-29T17:13:41.1142039+05:30	

Results		Messages					
	policy_id	policy_name	policy_type	pre_amount	duration_year	created_at	updated_at
1	101	Life Secure Plan	Life Insurance	12000	20	2023-12-01	2024-05-01
2	102	Health Plus Policy	Health Insurance	9350	1	2024-01-15	2024-06-20
3	103	Child Secure	Life Insurance	15000	18	2024-02-01	2024-06-01
4	104	Senior Health	Health Insurance	24200	1	2024-02-02	2024-06-02
5	105	Family Health Plus	Health Insurance	19800	2	2024-02-03	2024-06-03
6	106	Term Life Gold	Life Insurance	25000	30	2024-02-04	2024-06-04
7	107	Personal Accident	General Insurance	6000	1	2024-02-05	2024-06-05
8	108	Travel Secure	Travel Insurance	4000	1	2024-02-06	2024-06-06
9	109	Home Safe	Property Insurance	12000	10	2024-02-07	2024-06-07
10	110	Vehicle Protect	Motor Insurance	9000	1	2024-02-08	2024-06-08
11	111	Health Premium	Health Insurance	33000	3	2024-02-09	2024-06-09
12	112	Life Platinum	Life Insurance	35000	25	2024-02-10	2024-06-10

18. Delete PolicyAssignments whose EndDate is before today

DELETE FROM policy_assignments

WHERE end_date < GETDATE();

Messages	
Commands completed successfully.	
Completion time: 2025-12-29T17:15:08.1875358+05:30	

19. Number of rejected claims

```
SELECT COUNT(*) AS rejected_claims
FROM claims
WHERE claim_status = 'Rejected';
```

Results		Messages	
	rejected_claims		
1	2		

20. Computed premium fields

```
SELECT policy_id,
       policy_name,
       pre_amount,
       pre_amount * 0.06 AS local_taxes,
       pre_amount * 1.06 AS premium_with_tax,
       (pre_amount * 1.06) / 12 AS monthly_premium
FROM policies;
```

Results

Messages

	policy_id	policy_name	pre_amount	local_taxes	premium_with_tax	monthly_premium
1	101	Life Secure Plan	12000	720.00	12720.00	1060.000000
2	102	Health Plus Policy	9350	561.00	9911.00	825.916666
3	103	Child Secure	15000	900.00	15900.00	1325.000000
4	104	Senior Health	24200	1452.00	25652.00	2137.666666
5	105	Family Health Plus	19800	1188.00	20988.00	1749.000000
6	106	Term Life Gold	25000	1500.00	26500.00	2208.333333
7	107	Personal Accident	6000	360.00	6360.00	530.000000
8	108	Travel Secure	4000	240.00	4240.00	353.333333
9	109	Home Safe	12000	720.00	12720.00	1060.000000
10	110	Vehicle Protect	9000	540.00	9540.00	795.000000
11	111	Health Premium	33000	1980.00	34980.00	2915.000000
12	112	Life Platinum	35000	2100.00	37100.00	3091.666666

21. Add Address and City columns to Customers

```
ALTER TABLE insurance_customers
ADD address VARCHAR(200), city VARCHAR(50);
```

Messages	
Commands completed successfully.	
Completion time: 2025-12-29T17:15:08.1875358+05:30	

22. Add DevOfId column to Agents

ALTER TABLE agents

ADD DevOfId INT;

 Messages

Commands completed successfully.

Completion time: 2025-12-29T17:15:42.0851460+05:30

	agent_id	agent_name	phone	city	created_at	updated_at	DevOfId
1	5001	sanjay kumar	8919200290	Hyderabad	2023-11-10	2024-04-12	NULL
2	5002	monoj kumar	9123456781	delhi	2023-10-05	2024-03-18	NULL
3	5003	varshith	9988776651	Chennai	2023-09-20	2024-02-25	NULL
4	5004	Amit Rao	9011111111	Mumbai	2023-08-01	2024-04-01	NULL
5	5005	Priya Shah	9011111112	Pune	2023-08-02	2024-04-02	NULL
6	5006	Rakesh Jain	9011111113	Delhi	2023-08-03	2024-04-03	NULL
7	5007	Sunita Roy	9011111114	Kolkata	2023-08-04	2024-04-04	NULL
8	5008	Deepak Mishra	9011111115	Bhopal	2023-08-05	2024-04-05	NULL
9	5009	Anil Kumar	9011111116	Chennai	2023-08-06	2024-04-06	NULL
10	5010	Meena Iyer	9011111117	Coimbatore	2023-08-07	2024-04-07	NULL
11	5011	Suresh Naik	9011111118	Goa	2023-08-08	2024-04-08	NULL
12	5012	Kavya Shetty	9011111119	Mangalore	2023-08-09	2024-04-09	NULL
13	5013	Nitin Kulkarni	9011111120	Nagpur	2023-08-10	2024-04-10	NULL

23. Make DevOfId recursive foreign key

ALTER TABLE agents

ADD CONSTRAINT fk_dev_agent

FOREIGN KEY (DevOfId) REFERENCES agents(agent_id);

 Messages

Commands completed successfully.

Completion time: 2025-12-29T17:16:09.8898894+05:30

Results		Messages					
	agent_id	agent_name	phone	city	created_at	updated_at	DevOfId
1	5001	sanjay kumar	8919200290	Hyderabad	2023-11-10	2024-04-12	NULL
2	5002	monoj kumar	9123456781	delhi	2023-10-05	2024-03-18	NULL
3	5003	varshith	9988776651	Chennai	2023-09-20	2024-02-25	NULL
4	5004	Amit Rao	9011111111	Mumbai	2023-08-01	2024-04-01	NULL
5	5005	Priya Shah	9011111112	Pune	2023-08-02	2024-04-02	NULL
6	5006	Rakesh Jain	9011111113	Delhi	2023-08-03	2024-04-03	NULL
7	5007	Sunita Roy	9011111114	Kolkata	2023-08-04	2024-04-04	NULL
8	5008	Deepak Mishra	9011111115	Bhopal	2023-08-05	2024-04-05	NULL
9	5009	Anil Kumar	9011111116	Chennai	2023-08-06	2024-04-06	NULL
10	5010	Meena Iyer	9011111117	Coimbatore	2023-08-07	2024-04-07	NULL
11	5011	Suresh Naik	9011111118	Goa	2023-08-08	2024-04-08	NULL
12	5012	Kavya Shetty	9011111119	Mangalore	2023-08-09	2024-04-09	NULL
13	5013	Nitin Kulkarni	9011111120	Nagpur	2023-08-10	2024-04-10	NULL

24. List all policies for CustomerId = 5

```
SELECT p.*
FROM policies p
JOIN policy_assignments pa
ON p.policy_id = pa.policy_id
WHERE pa.customer_id = 5;
```

Results

Messages

	policy_id	policy_name	policy_type	pre_amount	duration_year	created_at	updated_at
1	105	Family Health Plus	Health Insurance	19800	2	2024-02-03	2024-06-03

25. View all customers with their policies

```
SELECT c.first_name, c.last_name, p.policy_name
FROM insurance_customers c
JOIN policy_assignments pa
ON c.customer_id = pa.customer_id
JOIN policies p
ON pa.policy_id = p.policy_id;
```

	first_name	last_name	policy_name
1	Sanjay	Attelli	Life Secure Plan
2	manoj	Kumar	Health Plus Policy
3	Amit	Sharma	Child Secure
4	Priya	Singh	Senior Health
5	Rahul	Verma	Family Health Plus
6	Neha	Gupta	Term Life Gold
7	Vikas	Reddy	Personal Accident
8	Pooja	Nair	Travel Secure
9	Arjun	Patel	Home Safe
10	Kiran	Das	Vehicle Protect
11	Sneha	Joshi	Health Premium
12	Rohit	Mehta	Life Platinum

26. View claims with customer name

```
SELECT c.first_name, c.last_name, cl.claims_amount
FROM insurance_customers c
JOIN policy_assignments pa
ON c.customer_id = pa.customer_id
JOIN claims cl
ON pa.assignment_id = cl.assignment_id;
```


	first_name	last_name	claims_amount
1	Sanjay	Attelli	50000
2	manoj	Kumar	12000
3	Amit	Sharma	20000
4	Priya	Singh	15000
5	Rahul	Verma	30000
6	Neha	Gupta	50000
7	Vikas	Reddy	8000
8	Pooja	Nair	12000
9	Arjun	Patel	25000
10	Kiran	Das	9000
11	Sneha	Joshi	40000
12	Rohit	Mehta	60000

27. FirstName, PolicyName, AgentName, StartDate, EndDate

```

SELECT c.first_name,
       p.policy_name,
       a.agent_name,
       pa.start_date,
       pa.end_date
FROM policy_assignments pa
JOIN insurance_customers c ON pa.customer_id = c.customer_id
JOIN policies p ON pa.policy_id = p.policy_id
JOIN agents a ON pa.agent_id = a.agent_id;

```

	first_name	policy_name	agent_name	start_date	end_date
1	Sanjay	Life Secure Plan	sanjay kumar	2024-01-01	2044-01-01
2	manoj	Health Plus Policy	monoj kumar	2024-06-15	2025-06-15
3	Amit	Child Secure	Amit Rao	2024-01-10	2042-01-10
4	Priya	Senior Health	Priya Shah	2024-02-01	2025-02-01
5	Rahul	Family Health Plus	Rakesh Jain	2024-03-01	2026-03-01
6	Neha	Term Life Gold	Sunita Roy	2024-04-01	2054-04-01
7	Vikas	Personal Accident	Deepak Mishra	2024-05-01	2025-05-01
8	Pooja	Travel Secure	Anil Kumar	2024-06-01	2025-06-01
9	Arjun	Home Safe	Meena Iyer	2024-07-01	2034-07-01
10	Kiran	Vehicle Protect	Suresh Naik	2024-08-01	2025-08-01
11	Sneha	Health Premium	Kavya Shetty	2024-09-01	2027-09-01
12	Rohit	Life Platinum	Nitin Kulkarni	2024-10-01	2049-10-01

28. Claims report with details

```
SELECT c.first_name,  
       p.policy_name,  
       cl.claims_amount,  
       cl.claim_status,  
       cl.claim_date  
FROM claims cl  
JOIN policy_assignments pa ON cl.assignment_id = pa.assignment_id  
JOIN insurance_customers c ON pa.customer_id = c.customer_id  
JOIN policies p ON pa.policy_id = p.policy_id;
```

	first_name	policy_name	claims_amount	claim_status	claim_date
1	Sanjay	Life Secure Plan	50000	Approved	2024-03-10
2	manoj	Health Plus Policy	12000	Pending	2024-08-05
3	Amit	Child Secure	20000	Approved	2024-04-15
4	Priya	Senior Health	15000	Pending	2024-05-10
5	Rahul	Family Health Plus	30000	Rejected	2024-06-05
6	Neha	Term Life Gold	50000	Approved	2024-07-01
7	Vikas	Personal Accident	8000	Pending	2024-08-12
8	Pooja	Travel Secure	12000	Approved	2024-09-18
9	Arjun	Home Safe	25000	Approved	2024-10-22
10	Kiran	Vehicle Protect	9000	Rejected	2024-11-05
11	Sneha	Health Premium	40000	Pending	2024-11-20
12	Rohit	Life Platinum	60000	Approved	2024-12-01

29. Customers with or without policies

```
SELECT c.*  
FROM insurance_customers c  
LEFT JOIN policy_assignments pa  
ON c.customer_id = pa.customer_id;
```

	customer_id	first_name	last_name	dob	phone	email	created_at	updated_at	address	city
1	1	Sanjay	Attelli	2005-01-29	8919200290	sanjay.attelli@gmail.com	2024-01-05	2024-06-10	NULL	NULL
2	2	manoj	Kumar	2004-05-25	9123456789	manoj.kumar@yahoo.com	2024-02-12	2024-07-01	NULL	NULL
3	3	Amit	Sharma	1998-04-12	9000000001	amit.sharma@gmail.com	2024-03-01	2024-06-01	NULL	NULL
4	4	Priya	Singh	1996-09-21	9000000002	priya.singh@gmail.com	2024-03-02	2024-06-02	NULL	NULL
5	5	Rahul	Verma	1994-01-15	9000000003	rahul.verma@yahoo.com	2024-03-03	2024-06-03	NULL	NULL
6	6	Neha	Gupta	2000-11-30	9000000004	neha.gupta@gmail.com	2024-03-04	2024-06-04	NULL	NULL
7	7	Vikas	Reddy	1992-07-18	9000000005	vikas.reddy@gmail.com	2024-03-05	2024-06-05	NULL	NULL
8	8	Pooja	Nair	1999-02-25	9000000006	pooja.nair@gmail.com	2024-03-06	2024-06-06	NULL	NULL
9	9	Arjun	Patel	1991-12-09	9000000007	arjun.patel@yahoo.com	2024-03-07	2024-06-07	NULL	NULL
10	10	Kiran	Das	1997-06-14	9000000008	kiran.das@gmail.com	2024-03-08	2024-06-08	NULL	NULL
11	11	Sneha	Joshi	1995-10-10	9000000009	sneha.joshi@gmail.com	2024-03-09	2024-06-09	NULL	NULL
12	12	Rohit	Mehta	1993-05-05	9000000010	rohit.mehta@gmail.com	2024-03-10	2024-06-10	NULL	NULL

30. Customers with NO claims

```
SELECT DISTINCT c.*
FROM insurance_customers c
LEFT JOIN policy_assignments pa ON c.customer_id = pa.customer_id
LEFT JOIN claims cl ON pa.assignment_id = cl.assignment_id
WHERE cl.claim_id IS NULL;
```

Results Messages									
customer_id	first_name	last_name	dob	phone	email	created_at	updated_at	address	city

31. Customer-wise total claim amount

```
SELECT c.first_name,
       SUM(cl.claims_amount) AS total_claim_amount
FROM insurance_customers c
JOIN policy_assignments pa ON c.customer_id = pa.customer_id
JOIN claims cl ON pa.assignment_id = cl.assignment_id
GROUP BY c.first_name;
```

Results Messages		
	first_name	total_claim_amount
1	Amit	20000
2	Arjun	25000
3	Kiran	9000
4	manoj	12000
5	Neha	50000
6	Pooja	12000
7	Priya	15000
8	Rahul	30000
9	Rohit	60000
10	Sanjay	50000
11	Sneha	40000
12	Vikas	8000

32. Agent-wise policy count

```
SELECT a.agent_name,
       COUNT(pa.policy_id) AS policy_count
FROM agents a
JOIN policy_assignments pa ON a.agent_id = pa.agent_id
GROUP BY a.agent_name;
```

Results Messages		
	agent_name	policy_count
1	Amit Rao	1
2	Anil Kumar	1
3	Deepak Mishra	1
4	Kavya Shetty	1
5	Meena Iyer	1
6	monoj kumar	1

